

Sean Luka Girgis

ETL / Data Warehouse QA Analyst | SQL, Python, Source-to-Target Validation & Data Quality

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Professional Summary

Senior ETL/data validation, data quality, and production technology professional with 20+ years of enterprise experience across SQL, Python, Pandas, ETL/ELT pipelines, data warehouse validation, source-to-target reconciliation, backend data testing, regression checks, troubleshooting, and stakeholder reporting. Strong background validating large operational datasets, writing SQL checks, comparing source and target outputs, investigating discrepancies, documenting defects, and automating repeatable validation workflows. Best aligned to ETL/Data Warehouse QA roles requiring complex SQL, Python/Pandas automation, row-count/null/duplicate/referential checks, data quality validation, regression testing, Agile collaboration, and production troubleshooting; Medicaid/HIPAA/claims 837/835, Informatica PowerCenter, Teradata, Azure DevOps Test Plans, HP ALM, SQLAlchemy, and healthcare data standards are positioned carefully as adjacent or learning areas rather than overstated production ownership.

Professional Experience

Senior Capacity & Data Engineer at CITI (2017-11 – 2025-12)

- Built and validated Python/Pandas/PySpark and SQL ETL pipelines ingesting telemetry from 6,000+ infrastructure endpoints into curated datasets for reporting, analytics, forecasting, and decision support.
- Designed source-to-target validation checks across operational feeds, Oracle datasets, AWS-backed reporting layers, and curated outputs to improve accuracy, completeness, and consistency.
- Wrote complex SQL queries across Oracle, Amazon Redshift, and reporting structures using joins, aggregations, filters, and reconciliation logic to validate transformation outputs and investigate discrepancies.
- Implemented data quality checks for row counts, null rates, duplicate keys, unmatched joins, referential-style consistency, freshness, abnormal deltas, and output comparisons against trusted prior runs.
- Troubleshot production data issues by tracing source feeds, SQL transformation logic, Python/Pandas processing, missing or late data, failed outputs, latency symptoms, and downstream reporting discrepancies.
- Documented validation logic, source mappings, metric definitions, data flow notes, support procedures, runbooks, and issue findings to improve repeatability and auditability.
- Partnered with infrastructure, application, analytics, engineering, and business stakeholders to clarify data requirements, explain defects, validate fixes, and improve reporting reliability.

Performance Engineer at G6 Hospitality LLC (2017-03 – 2017-11)

- Analyzed and validated Dynatrace AppMon and Gomez Synthetic Monitoring datasets for Brand.com and critical hospitality systems.
- Investigated data quality issues, performance trends, missing signals, and reporting anomalies in real-user and synthetic monitoring outputs.
- Created dashboards and reporting views that converted raw monitoring data into actionable insights for business and engineering stakeholders.

SME for CA APM (Senior Consultant) at CA Technologies / TIAA-CREF (2011 – 2016)

- Supported a large financial-services monitoring and reporting environment with 50+ Enterprise Managers and 4,000–6,000 instrumented agents.
- Built Perl and KornShell automation for extracting, profiling, validating, transforming, and distributing APM telemetry, replacing manual reporting workflows with repeatable data validation processes.
- Designed dashboards, alert rules, SLA thresholds, reporting standards, and technical documentation used by support, infrastructure, and application teams.

Performance Test Engineer at AT&T (2010-08 – 2011-07)

- Developed and executed performance and regression test plans for enterprise telecom applications, including baseline comparisons and release-readiness analysis.
- Analyzed application behavior under load to identify throughput limits, SQL/JDBC bottlenecks, CPU, memory, thread, and garbage-collection issues.
- Documented test results, expected behavior, defects, risks, and remediation recommendations for engineering and release teams.

Senior Systems & Data Migration Engineer at Sabre (2008-05 – 2010)

- Led migration validation for a high-throughput shopping engine moving from 200+ MySQL nodes to a 6-node Oracle RAC cluster.
- Built C++/OCCI/OCI validation and regression testing utilities to verify correctness across millions of migrated transaction records.
- Optimized SQL and Oracle access patterns for high-volume workloads while reducing physical hardware footprint by 95%.

Architect / Developer (IRS CADE Project) at Computer Science Corporation (CSC) (2007-10 – 2008-05)

- Designed UML class and sequence diagrams and developed modules for IRS CADE mainframe modernization.
- Built CICS, MQ Series, XML messaging interfaces, and DB2-backed components in a government modernization environment.

Developer / Support Engineer (Billing, Interfaces, CSM/PRMS) at Corpus Inc. / Sprint (2001 – 2007)

- Developed and supported billing, customer-management, and financial-style operational interfaces using Oracle, PL/SQL, C/C++, Pro*C, SQL scripts, XML, and messaging patterns.
- Built automation scripts in KornShell/ksh, Perl, Awk, Syncsort, and SQL to support data extraction, transformation, reporting, deployments, validation, and production housekeeping.
- Provided production support, defect investigation, data maintenance, issue escalation, and troubleshooting for telecom billing and customer-management applications.

Education

Post-Graduate Diploma in Computer Engineering Technology / Computer Science

Humber College, Toronto, Canada

Bachelor of Science in Civil Engineering

Zagazig University, Egypt

Skills

Flagship Projects

Production Data Quality and RCA Workflow

Built validation and reconciliation workflows for production reporting datasets, including row-count checks, null checks, duplicate/unmatched record checks, missing feed diagnosis, and prior-run comparisons.

Technologies: SQL, Python, Pandas, Data Validation, RCA, Runbooks

Serverless Lakehouse Platform (AWS)

Designed an AWS data platform pattern using S3, Glue, Athena, Redshift, PySpark, and Parquet with validation checkpoints, SQL quality checks, and curated analytics datasets.

Technologies: AWS S3, AWS Glue, Athena, Redshift, PySpark, SQL, Data Quality

Enterprise APM Reporting Automation

Built Perl and KornShell automation for extracting, profiling, validating, transforming, and distributing performance telemetry across large financial-services monitoring environments.

Technologies: KornShell/ksh, Perl, Unix/Linux, CA APM, SQL, Automation